Keheliya Gallaba

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Google Scholar Profile: https://scholar.google.ca/citations?user=FF_M_CEAAAAJ&hl=en
DBLP: https://dblp.uni-trier.de/pers/hd/g/Gallaba:Keheliya

SUMMARY

- Applied AI researcher and engineer with **14+ years** of hands-on experience; currently helping build a **software engineering foundation model from scratch** and its evaluation stack at Huawei's Centre for Software Excellence.
- Deep, practical expertise in **agentic architectures, RAG engineering, and LLM evaluation & benchmarking**; translated research into **4 patent filings** on foundation model-powered software.
- Experienced **public speaker and educator**: publicly available lecture recordings on AI agents and RAG, graduate teaching (Queen's University), AIware Bootcamps (Toronto, Ottawa, Amsterdam), and tutorials at **KDD'25** & **FSE'25**.
- **Client-facing delivery** experience: Solution development, deployment, and on-site support for enterprise customers in the US, Switzerland, Sweden, and New Zealand (WSO2); developer tools at **Microsoft Research, Fortinet** & **Wise Systems**.
- Published in top-tier (**CORE Rank: A* & A**) Software Engineering venues (**ICSE, TSE, FSE, ASE**); ACM SIGSOFT Distinguished Paper Award at **ASE '25**, best paper award at **ESEM '15**.
- Mentored 12 students to publish; successful collaborations with distributed teams in **Canada, Japan, Austria,** and **USA**.
- Canadian/Sri Lankan dual citizen.

SKILLS

AI/LLM: Agentic architectures & orchestration (single/multi-agent, tool use, planning), RAG systems, prompt engineering, LLM evaluation & benchmark harness engineering.
Programming: Python, Java, JavaScript/Node.js, R.
Tools & Platforms: Git, Docker, CI/CD platforms (CircleCI, Travis CI), cloud-based build & evaluation infrastructure.

CURRENT EMPLOYMENT

Waterloo Centre for Software Excellence, Huawei Canada, 1055 Princess St., Kingston, ON, Canada

Staff Researcher, Foundation Models & Software Engineering Group — Jun 2021 – Present

Supervisor: Prof. Ahmed E. Hassan

Core contributor to building a **software engineering foundation model from scratch**, owning the evaluation side: benchmarking the coding capabilities of foundation model agents, and validating the benchmarks themselves before trusting their scores.

Co-led a "benchmarking the benchmarks" effort: introduced Oracle-run (gold-patch) validation for agentic coding benchmarks such as SWE-bench and feature-implementation benchmarks, diagnosed harness and environment instability, and released **FeatBench-Verified v1.3** (155 tasks), raising gold-patch resolution from 37% to 99.8%. [**link**]

Researching **trustworthy coding agents** and agentic program repair; built and published agentic workflow-generation and multi-agent requirements-refinement approaches; translated research into **4 patent filings** on foundation model-powered software.

Designed and delivered public lectures, tutorials, and bootcamps on **AI agents and RAG engineering** for industry and academic audiences (see Public Lectures below); co-organized international conferences and workshops to foster industry–academia collaboration.

INDUSTRIAL EXPERIENCE

Wise Systems Inc., Montréal, QC, Canada

Senior Software Engineer — Aug 2020 – Apr 2021

Developing features for AI-driven dispatch and routing software that enable last mile delivery, improving fleet efficiency and customer service through machine learning and real-time data.

Fortinet Inc., Vancouver, BC, Canada

Software Development Engineer II, Forticlient Team — Feb 2016 – Aug 2016

Worked on the system for generating software vulnerability signatures, validating them, and delivering them to endpoint security software (Forticlient).

WSO2 Inc., Colombo, Sri Lanka

Senior Software Engineer, Integration Technologies Team — Apr 2013 – Nov 2013

Development and support of open source middleware solutions for enterprise integration problems.
Client-facing delivery: PoC development, deployment, and on-site support for enterprise customers in the US, Switzerland, Sweden, and New Zealand; worked directly with customer teams to translate their requirements into working integration solutions.

Software Engineer, Integration Technologies Team — Dec 2011 – Mar 2013
Worked with WSO2 Business Process Server, Enterprise Service Bus, Identity Server & Governance Registry.
Experience with WS-BPEL & other WS-* standards, SOAP/REST web services.
XPATH, XSLT, Apache ODE and Eclipse plug-in development.

RESEARCH EXPERIENCE

YourBase Inc., Redmond, WA, USA
Visiting Researcher — Jun 2019 – Jul 2020
Supervisors: Yves Junqueira and John Ewart
Focus: Accelerating continuous integration by caching environments and inferring dependencies.

Microsoft Research, Cambridge, UK
PhD Research Intern, Systems and Networking Group — Jun 2018 – Sep 2018
Supervisors: Dr Katja Kevic and Dr Brendan Murphy
Focus: Automatic removal of feature gates in source code.

PUBLIC LECTURES & INVITED TALKS

Agentic Architectures and Workflows — Lecture, 90 mins [**video**]
First delivered in the **CISC877: Software Engineering and Foundation Models** graduate course at Queen's University; variants later presented at AIware Bootcamps in Toronto and Ottawa.
Covers agent fundamentals (perception, memory, tool use with Toolformer/ReAct), planning and theory of mind, cognitive architectures, agent patterns (chains, routers, graphs), static vs. dynamic orchestration, multi-agent collaboration and self-reflection, and agents in software engineering.

RAG Engineering — Lecture, 90 mins [**video**]
First delivered in **CISC877** at Queen's University; variants later presented at AIware Bootcamps in Toronto and Ottawa.
Covers indexing, retrieval, and generation; query translation (multi-query, RAG-fusion, least-to-most, step-back prompting, HyDE); logical and semantic routing; advanced indexing (multi-representation, RAPTOR, ColBERT); Graph RAG; re-rankers; large-context challenges and context caching; and a reference architecture for enterprise-grade, production-ready RAGware.

Beyond Vibe Coding: Leveraging Foundation Models to Infer Intent — Invited Talk, 30 mins [**video**]
First delivered at the **RAISE 2025** workshop (Requirements engineering for AI-powered SoftwarE) co-located with ICSE'25 in Ottawa; a variant later delivered at **AIware Bootcamp Europe 2025** in Amsterdam, co-located with OSS Europe 2025.
Covers inferring user intent with foundation models: Theory of Mind and iterative refinement, conversational interfaces for requirements elicitation, rubrics for evaluating intent quality, real-world gains over baselines, and open challenges (sycophancy, hallucination, compute cost).

TEACHING EXPERIENCE

Visiting Instructor

Queen's University, Kingston, ON, Canada
CISC877 Software Engineering and Foundation Models — Fall 2023 & Winter 2025

University of Moratuwa, Sri Lanka
CS2212 Programming Challenge II — May 2011 – Sep 2011
CS2202 Programming Challenge — Sep 2010 – Dec 2010

Teaching Assistant

McGill University, Montréal, QC, Canada
ECSE321 Introduction to Software Engineering — Fall 2017

University of British Columbia, Vancouver, BC, Canada
EECE416 Software Testing and Analysis — Winter II 2014
EECE310 Software Engineering — Winter II 2013 & Winter I 2014

Curriculum Committee Member & Instructor

AIware Bootcamp Mini 2025, Ottawa, ON, Canada [**link**] — May 2025
Co-located with ICSE'25, 50 attendees

AIware Leadership Bootcamp 2024, Toronto, ON, Canada [**link**] — Nov 2024
80 attendees (Primarily graduate students and SE faculty)

Curriculum Committee Member

The Hitchhiker's Guide to Production-ready Trustworthy Foundation Model-Powered Software (FMware) [**link**] Jul 2025
3-hour lecture-style tutorial at KDD'25, Toronto, Canada

A Tutorial on Software Engineering for FMware [**link**] Jun 2025
1.5-hour tutorial at FSE'25, Trondheim, Norway

PATENTS

[P1] G. K. Rajbahadur, D. Lin, K. Gallaba, F. R. Cogo, and A. E. Hassan, "Methods and Systems for Augmenting Synthetic Data Labels with Human-Generated Data Labels", US Patent Application US-2026105313-A1, Published Apr 2026.

[P2] K. Gallaba, A. Arabat, D. Lin, and A. E. Hassan, "Data Generation Methods and Apparatuses, and Related Products", International Patent Application (PCT) WO-2026076802-A1, Assignee: Huawei Technologies Co., Ltd., Published Apr 2026.

[P3] K. Gallaba, F. R. Cogo, D. Lin, G. K. Rajbahadur, and A. E. Hassan, "Hierarchical Dynamic Planning of Foundation Model Agents", International Patent Application (PCT) WO-2026066496-A1, Assignee: Huawei Technologies Co., Ltd., Published Apr 2026.

[P4] F. R. Cogo, D. Lin, J. Lin, K. Gallaba, G. K. Rajbahadur, and A. E. Hassan, "Machine Learning Asset Management", International Patent Application (PCT) WO-2026016694-A1, Assignee: Huawei Technologies Co., Ltd., Published Jan 2026.

EDUCATION

McGill University, Montréal, QC, Canada

Ph.D. in Software Engineering Sep 2016 – Nov 2021
Thesis: Improving the Robustness and Efficiency of Continuous Integration and Continuous Deployment
Advisor: Prof. Shane McIntosh
Publications: MSR17a, MSR17b, MSR18a, MSR18b, TSE18, ASE18, MSR19, ICSME19, TSE20, ICSE22.

University of British Columbia, Vancouver, BC, Canada

M.A.Sc. in Software Engineering Jan 2014 – Dec 2015
Thesis: Characterizing and refactoring asynchronous JavaScript callbacks
Advisors: Prof. Ali Mesbah and Prof. Ivan Beschastnikh
Publications: ESEM15, ICSME17.

University of Moratuwa, Moratuwa, Sri Lanka

B.Sc. (Hons.) in Computer Science and Engineering May 2007 – Nov 2011
Thesis: An analysis of BPMN 2.0 as a business process execution language
Advisors: Vishaka Nanayakkara and Dr Milinda Pathirage
Publications: CeBPM12.

AWARDS & RECOGNITION

ACM SIGSOFT Distinguished Paper Award Nov 2025
Awarded at the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE '25).

Mitacs Accelerate International Award and Graduate Mobility Award May 2019
Funding to study and conduct research abroad as part of the McGill degree program. Award Amount: $6,700

FRQNT Doctoral Research Award (Bourses de doctorat en recherche) May 2018
Ranked #1 in the ICT group and obtained the competitive scholarship provided by Fonds de recherche du Québec – Nature et technologies (FRQNT). Award Amount: $28,000

McGill Engineering Doctoral Award (MEDA) 2016 – 2019
Full scholarship with stipend for PhD studies at McGill University. Award Amount: $96,000

Best Full Paper Award Oct 2015
Awarded at The ACM/IEEE International Symposium on Empirical Software Engineering and Measurement.

ISERN Student Travel Grant Oct 2015
Awarded by The International Software Engineering Research Network (ISERN) to attend ESEM '15.

Manthan Award South Asia for e-Localization Dec 2010
For creating tools that allowed viewing non-Unicode Sinhala language content on Unicode-enabled computers.

e-Swabhimani 2010 award for e-Localization Nov 2010
Awarded at the National Best e-Content Awards, Sri Lanka.

COMPLETE LIST OF PUBLICATIONS

FULL PAPERS IN INTERNATIONAL JOURNALS (CORE RANK: A*)

[TOSEM26b] N. Nashid, D. Ding, K. Gallaba, A. E. Hassan, and A. Mesbah, "Beyond Accuracy: Behavioral Dynamics of Agentic Multi-Hunk Repair", *ACM Transactions on Software Engineering and Methodology (TOSEM)*, Just Published, Sep 2026. [**PDF**]

[TOSEM26a] S. Masoumzadeh, K. Gallaba, D. Lin, and A. E. Hassan, "Towards Reliable Generation of Executable Workflows by Foundation Models," *ACM Transactions on Software Engineering and Methodology (TOSEM)*, Just Published, Apr 2026. [**PDF**]

[TSE20] K. Gallaba, Y. Junqueira, J. Ewart, and S. McIntosh, "Accelerating Continuous Integration by Caching Environments and Inferring Dependencies," *IEEE Transactions on Software Engineering*, 48(6): pp. 2040–2052, 2020. [**PDF**]

[TSE18] K. Gallaba and S. McIntosh, "Use and Misuse of Continuous Integration Features: An Empirical Study of Projects that (mis)use Travis CI," *IEEE Transactions on Software Engineering*, 46(1): pp. 33–50, 2018. [**PDF**]

FULL PAPERS IN INTERNATIONAL CONFERENCES (CORE RANK: A* & A)

[ICSE26] G. K. Rajbahadur, K. Gallaba, E. Rashno, A. Suriyawongkul, K. Bennet, K. Stewart, and A. E. Hassan, "Building an Open AIBOM Standard in the Wild", In *Proceedings of the International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)*, 842–853, Apr 2026. [**PDF**]

[ASE25] N. Nashid, D. Ding, K. Gallaba, A. E. Hassan, and A. Mesbah, "Characterizing Multi-Hunk Patches: Divergence, Proximity, and LLM Repair Challenges", In *Proceedings of the International Conference on Automated Software Engineering (ASE)*, 1629–1641, Nov 2025. **ACM SIGSOFT Distinguished Paper Award** [**PDF**]

[KDD25] K. Vasilevski, B. Rombaut, G. K. Rajbahadur, G. A. Oliva, K. Gallaba, F. R. Cogo, D. Lin, H. Zhang, B. Chen, K. Thangarajah, A. E. Hassan, Z. M. Jiang, "The Hitchhikers Guide to Production-ready Trustworthy Foundation Model powered Software (FMware)", In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 6162–6172, Jul 2025. [**PDF**]

[ASE24] M. Yin, Y. Kashiwa, K. Gallaba, M. Alfadel, Y. Kamei, and S. McIntosh, "Developer-Applied Accelerations in Continuous Integration: A Detection Approach and Catalog of Patterns", In *Proceedings of the International Conference on Automated Software Engineering (ASE)*, 1655–1666, Oct 2024. [**PDF**]

[FSE24] A. E. Hassan, D. Lin, G. K. Rajbahadur, K. Gallaba, F. R. Cogo, B. Chen, H. Zhang, K. Thangarajah, G. A. Oliva, J. Lin, W. M. Abdullah, Z. M. Jiang, "Rethinking Software Engineering in the Foundation Model Era", In *Companion Proceedings of the International Conference on the Foundations of Software Engineering (FSE)*, 294–305, Jul 2024. [**PDF**]

[ICSE22] K. Gallaba, M. Lamothe, and S. McIntosh, "Lessons from Eight Years of Operational Data from a Continuous Integration Service: A Case Study of CircleCI", In *Proceedings of the International Conference on Software Engineering (ICSE)*, 1330–1342, May 2022. **Nominated for ACM SIGSOFT Distinguished Paper Award** [**PDF**]

[ASE18] K. Gallaba, C. Macho, M. Pinzger, and S. McIntosh, "Noise and Heterogeneity in Historical Build Data: An Empirical Study of Travis CI", In *Proceedings of the International Conference on Automated Software Engineering (ASE)*, 87–97, Sep 2018. [**PDF**]

[ICSME17] K. Gallaba, Q. Hanam, A. Mesbah, and I. Beschastnikh, "Refactoring Asynchrony in JavaScript", In *Proceedings of the International Conference on Software Maintenance and Evolution (ICSME)*, 353–363, Sep 2017. [**PDF**]

[ESEM15] K. Gallaba, A. Mesbah, and I. Beschastnikh, "Don't Call Us, We'll Call You: Characterizing Callbacks in JavaScript", In *Proceedings of the International Symposium on Empirical Software Engineering and Measurement (ESEM)*, pp. 247–256, Oct 2015. **Best Paper Award** [**PDF**]

SHORT PAPERS IN INTERNATIONAL CONFERENCES (CORE RANK: A)

[ICSME19] K. Gallaba, "Improving the Robustness and Efficiency of Continuous Integration and Deployment", In *Proceedings of the International Conference on Software Maintenance and Evolution (ICSME)*, pp. 619–623, Sep 2019. [**PDF**]

[MSR19] D. Abric, O. E. Clark, M. Caminiti, K. Gallaba, and S. McIntosh, "Can Duplicate Questions on Stack Overflow Benefit the Software Development Community?", In *Proceedings of the 16th International Conference on Mining Software Repositories (MSR), Mining challenge*, pp. 230–234, May 2019. [**PDF**]

[MSR18a] N. Rabbani, M. S. Harvey, S. Saquif, K. Gallaba, and S. McIntosh, "Revisiting "Programmers' Build Errors" in the Visual Studio Context", In *Proceedings of the 15th International Conference on Mining Software Repositories (MSR), Mining challenge*, pp. 98–101, May 2018. [**PDF**]

[MSR18b] R. Amlekar, A. F. R. Gamboa, K. Gallaba, and S. McIntosh, "Do Software Engineers Use Autocompletion Features Differently Than Other Developers?", In *Proceedings of the 15th International Conference on Mining Software Repositories (MSR), Mining challenge*, pp. 86–89, May 2018. [**PDF**]

[MSR17a] M. Manglaviti, E. Coronado-Montoya, K. Gallaba, and S. McIntosh, "An Empirical Study of the Personnel Overhead of Continuous Integration", In *Proceedings of the 14th International Conference on Mining Software Repositories (MSR), Mining challenge*, pp. 471–474, May 2017. [**PDF**]

[MSR17b] Y. Khan, Y. Gupta, K. Gallaba, and S. McIntosh, "The Impact of the Adoption of Continuous Integration on Developer Attraction and Retention", In *Proceedings of the 14th International Conference on Mining Software Repositories (MSR), Mining challenge*, pp. 491–494, May 2017. [**PDF**]

ONGOING RESEARCH

[arXiv] R. Shayanfar, K. Gallaba, and A. E. Hassan, "What Does an Agentic Software Engineering Benchmark Measure? Profiling Task Demands and Agent Behaviour Beyond What Category Labels Reveal", Under Review, Sep 2026. [**PDF**]

[arXiv] B. A. Muse, L. Briand, Y. Lu, and K. Gallaba, "CASPER—Change-Aware Slice Prioritization for Efficient Regression Testing of LLM-based Systems", Under Review, Aug 2026. [**PDF**]

[arXiv] A. Ajibode, O. Ben Sghaier, K. Gallaba, B. Adams, and A. E. Hassan, "Towards Imputation of Pre-Trained Language Model Metadata using Semantic Fingerprinting", Under Review, Jun 2026. [**PDF**]

[arXiv] K. Gallaba, A. Arabat, D. Lin, M. Sayagh, and A. E. Hassan, "Towards Conversational Development Environments: Using Theory-of-Mind and Multi-Agent Architectures for Requirements Refinement", Under Review, May 2025. [**PDF**]

PROFESSIONAL SERVICE

Organizing committees: AIware 2024 (Proceedings Chair); SCAM 2021 (Social Media Co-chair).

Program committees: ICSE (Research Track 2027; SEIP Track 2024–2026), AGENT 2026, MSR (Data & Tools 2025–2026; Industry 2023, 2025; Junior PC 2023), ICSME (Industry 2024–2025), SANER 2024, FORGE 2024, ICPC (Tools 2019), OOPSLA (Artifact Evaluation 2017).

Journal reviewer: TOSEM, TSE, EMSE, JSS, IST, IEEE Software.

INVITED EVENTS

FM+SE Vision 2030 — Nov 2023
A forum for industrial and academic experts to discuss advances at the intersection of Software Engineering and Foundation Models, held in Mexico City, Mexico.

Shonan Meeting No. 152 — Dec 2019
on "Release Engineering for Mobile Applications" held in Kanagawa, Japan.

Shonan Meeting No. 095 — Mar 2017
on "Mining Software Repositories: Accomplishments, Challenges and Future Trends" held in Kanagawa, Japan.

MENTORING EXPERIENCE

Undergraduate Students (Co-advised with Dr. Shane McIntosh)

Durham Abric, Oliver Clark, Matthew Caminiti — Sep 2018 – Apr 2019
Co-authored Papers: MSR19

Rahul Amlekar, Noam Rabbani, Mike Harvey, Sadnan Saquif, Andrés Felipe Rincón Gamboa — Sep 2017 – Apr 2018
Co-authored Papers: MSR18a, MSR18b

Marco Manglaviti, Eduardo Coronado-Montoya, Yash Gupta, Yusaira Khan — Sep 2016 – Apr 2017
Co-authored Papers: MSR17a, MSR17b

OPEN SOURCE CONTRIBUTIONS

Gephi, Open source graph visualization and manipulation software

System Developer/GSoC Student — Jan 2011 – May 2011
Developed an automated Maven build to fasten the release process of Gephi. [**Team Profile**] [**Details**]

Crawljax, a tool for automatically crawling and testing modern web applications

Contributor/Maintainer — Nov 2013 – Aug 2014
Fixing bugs, updating dependencies, and other routine maintenance. [**Project**]

Other,

Localization Contributor — Jan 2008 – Present
Localization of FOSS projects including Ubuntu, Sugar Labs, GNOME and Miro [**Launchpad**]

Minor Contributor — Feb 2011 – Present
Contributed via patches and pull requests to well-known OSS projects. [**GitHub**]

MEMBERSHIPS

Professional Member Association for Computer Machinery (ACM)
Member ACM Special Interest Group on Software Engineering (SIGSOFT)

REFERENCES

Professor Shane McIntosh (Relationship: PhD Thesis Advisor)
Associate Professor and Associate Director of Graduate Studies
David R. Cheriton School of Computer Science, University of Waterloo
Waterloo, ON, Canada N2L 3G1
shane.mcintosh@uwaterloo.ca

Professor Ahmed E. Hassan
Professor and Canada Research Chair
School of Computing, Queen's University
Kingston, ON, Canada K7L 1A8
ahmed@cs.queensu.ca

Professor Ali Mesbah (Relationship: MASc Thesis Advisor)
Professor of Software Engineering
Department of Electrical and Computer Engineering, University of British Columbia (UBC)
KAIS 4044, 2332 Main Mall, Vancouver, BC, Canada V6T 1Z4
amesbah@ece.ubc.ca

Professor Massimiliano Di Penta
Professor of Computer Science
University of Sannio
Via Traiano, 9 - 82100 Benevento, Italy
dipenta@unisannio.it